

Labor 2,

Bài 1, unigram

1,

$V = [\text{she}, \text{cat}, \text{eats}, \text{fish}, \text{likes}, \text{dog}, \text{meat}]$

~~$\Rightarrow N = 7$~~

2) $N = 12$.

3,

$$P(\text{she}) = \frac{C(\text{she})}{N} = \frac{3}{12} = \frac{1}{4}$$

$$P(\text{cat}) = \frac{2}{12} = \frac{1}{6}$$

$$P(\text{fish}) = \frac{2}{12} = \frac{1}{6}$$

$$P(\text{dog}) = \frac{1}{12}$$

4,

$$\sum P(w) = P(\text{she}) + P(\text{cat}) + P(\text{fish}) + P(\text{dog}) \\ + P(\text{likes}) + P(\text{meat}) + P(\text{eats})$$

$$= \frac{1}{4} + \frac{1}{6} + \frac{1}{6} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{6}$$

$$= 1$$

Bài 2,

No.

$$p(\text{cat} | \text{she}) = \frac{C(\text{she cat})}{C(\text{she})} = \frac{2}{3}$$

$$p(\text{dog} | \text{she}) = \frac{C(\text{she dog})}{C(\text{she})} = \frac{1}{3}$$

$$p(\text{cats} | \text{cat}) = \frac{C(\text{cat cats})}{C(\text{cat})} = \frac{1}{2}$$

$$p(\text{likes} | \text{cat}) = \frac{C(\text{cat likes})}{C(\text{cat})} = \frac{1}{2}$$

~~$$p(\text{cat}) = \frac{1}{6}, p(\text{dog}) = \frac{1}{12}$$~~

Tổng các xác suất sau she là $\sum_{w \in V} p(w | \text{she}) = p(\text{cat} | \text{she}) + p(\text{dog} | \text{she}) + p(\text{likes} | \text{she}) + p(\text{cats} | \text{she}) + p(\text{meats})$

$$= \frac{2}{3} + \frac{1}{3} = 1$$

Tổng các xác suất sau "she" là.

$$\sum_{w \in V} p(w | \text{she}) = p(\text{cat} | \text{she}) + p(\text{dog} | \text{she}) + p(\text{likes} | \text{she}) + p(\text{cats} | \text{she}) + p(\text{meats} | \text{she}) + p(\text{fish} | \text{she})$$

$$= \frac{1}{3} + \frac{2}{3} + 0 + 0 + 0 + 0 = 1$$

Bài 11: $p(\text{fish} | \text{cats}) = \frac{p(\text{cats fish})}{p(\text{cats})} = \frac{1}{2}$

$$= p(\text{she cat eats fish})$$

$$= \frac{1}{4} \cdot \frac{2}{3} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{2}{48} = \frac{1}{24}$$

- Nếu thêm 1 từ vào câu, xác suất của cả câu bằng 0 vì do nhân với số < 1 nên kết quả cuối cùng sẽ giảm

Bài 4,

S_1 : she cat eats fish

$$p(S_1) = p(\text{fish}) p(\text{she}) p(\text{she} | \text{cat})$$

$$= p(\text{she})$$

~~S_1 : she cat eats fish.~~

Bigram.

$$p(S_1) = \frac{1}{24} \approx 0,041.$$

$$p(S_1) = p(\text{she}) p(\text{cat} | \text{she}) p(\text{eats} | \text{cat}) p(\text{fish} | \text{eats})$$

$$= \frac{1}{4} \cdot \frac{1}{3} \cdot 1 \cdot \frac{1}{2}$$

$$= \frac{1}{24} \approx 0,041 \Rightarrow S_1 \text{ có xác suất xuất hiện } = S_1$$

Trigram

$$p(S_1) = p(\text{she}) p(\text{cat} | \text{she}) p(\text{eats} | \text{she cat}) p(\text{fish} | \text{cat eats})$$

$$= \frac{1}{4} \cdot \frac{2}{3} \cdot \frac{1}{2} = 1$$

$$= \frac{1}{12} \approx 0,083$$

$$p(S_2) = p(\text{she}) p(\text{dog} | \text{she}) p(\text{eats} | \text{she dog}) p(\text{fish} | \text{dog eats})$$

$$= \frac{1}{4} \cdot \frac{1}{3} \cdot 1 \cdot 0 = 0$$

$\Rightarrow$ S_1 có xác suất xuất hiện Cao hơn S_2